INDIAN STOCK MARKET ANALYSIS
Final Insights & Conclusion

Python • Pandas • MySQL • Seaborn | Notebook-based findings

1. Executive Summary

This report summarizes the results shown in Displaying_stocks.ipynb. The notebook contains analysis of 15 Indian equity tickers across nine sectors, including closing-price maxima, top daily gains, cumulative traded value, sector composition, average daily returns, and a stock-performance SQL view. All reported figures are taken from saved notebook outputs, not independently verified against exchange records.

2. Dataset and Analytical Scope

Coverage in displayed results: 15 stocks and 9 sectors. The top-gain results include dates from January 2025 through July 2026, but the complete dataset start/end dates and record count cannot be established from the notebook outputs alone. The notebook uses a MySQL stock_market table and retrieves aggregated results with Pandas for visualization.

3. Key Findings

Metric	Leader / Value	Interpretation
Highest observed closing price	MARUTI ₹18,762.97	This is a nominal share-price maximum, not evidence of the highest investment return.
Largest single-day percentage gain	MARUTI 7.76% on 7 Aug 2025	Other prominent daily gains: KOTAKBANK 6.51% (9 Jul 2026), RELIANCE 6.10% (17 Jan 2025).
Highest cumulative traded value	MARUTI approximately 6.446764×10^{13} (source units)	TITAN ranked second (2.383653×10^{13}), followed by TCS (1.847287×10^{13}).
Highest average daily return	RELIANCE 0.21%	Followed by TITAN 0.17% and AXISBANK 0.15%; these are arithmetic averages of daily percentage returns.
Largest sector by company count	Banking 5 companies	IT and FMCG have 2 companies each; six other

sectors have 1 each.

4. Sector and Return Insights

The saved sector-level stock return output shows RELIANCE (Energy) at 0.21%, TITAN (Consumer) at 0.17%, and AXISBANK (Banking) at 0.15% average daily return. Within IT, TCS (0.09%) exceeded INFY (0.06%); within FMCG, ITC (0.03%) exceeded HINDUNILVR (0.02%). SUNPHARMA recorded −0.12%, and BHARTIARTL rounded to 0.00%.

The sector rank query orders AVG(daily_return_pct) in ascending order. Accordingly, its rnk=1 identifies the lowest, not highest, average daily return within each sector. The heatmap values remain useful, but any top-performer interpretation should use descending rank or compare the actual return values.

5. Visual Interpretation

Highest closing price bar chart

Shows differences in maximum nominal share prices. It should not be described as a return-performance ranking.

Top daily gain bar chart

Highlights exceptional single-day upward moves, not sustained performance or expected future gains.

Cumulative traded value chart

Compares the sum of market_value_traded over available observations; totals depend on data coverage and the field's unit definition.

Sector composition chart

Banking has the greatest representation among the 15 analyzed tickers, so cross-sector comparisons may reflect unequal sample sizes.

Average daily return heatmap

Highlights cross-stock variation in mean daily percentage returns. Small differences and rounding should be interpreted cautiously.

Stock performance bubble chart

Combines average close, mean daily return, and cumulative volume. Bubble size measures volume rather than investment performance.

6. Limitations and Quality Checks

- Only 15 tickers appear in the saved outputs; this is not a complete NIFTY 50 constituent analysis.
- No data provenance, full observation date range, sample size per ticker, or adjustment policy is documented in the notebook.
- Cumulative traded value and total volume may be distorted if stocks have unequal observation counts.
- Average daily return is not equivalent to compounded period return; the report makes no forward-looking investment claims.
- The notebook includes Pandas DBAPI connection warnings and Seaborn palette deprecation warnings. These do not automatically invalidate query results, but can be cleaned up for reproducibility.
- The saved notebook includes database credentials in plain text; remove them and use environment variables before publishing the notebook publicly.

7. Final Conclusion

The project demonstrates a practical workflow from structured stock-market data in MySQL to SQL-based aggregation, Pandas result retrieval, and Seaborn visualization. Among the 15 tickers in the notebook, Maruti Suzuki stands out for its highest observed nominal closing price, largest listed single-day percentage gain, and largest cumulative traded value; Reliance leads the displayed average daily return metric. Banking is the most represented sector in this sample. Together, these findings reveal how different metrics describe different aspects of historical stock behavior—price level, daily movement, trading activity, and mean return—without implying that one metric alone identifies the best investment.

8. Recommended Next Steps

For a portfolio-ready submission, document the source and date coverage, include row counts and missing-value checks, fix sector ranking to DESC when seeking top performers, and attach selected charts with captions. Add the SQL scripts and a README describing methodology and limitations. Consider a later AWS S3 / Snowflake extension as a separate data-engineering enhancement.

Source: Displaying_stocks.ipynb (uploaded notebook; saved query outputs).